

A Low-Latency Digital Servo for Optical Frequency Comb Stabilization

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As a fundamental tool in precision spectroscopy and metrology, the optical frequency comb demands exceptional stability and accuracy across diverse applications. Digital servo systems, offering advantages such as low cost and high flexibility, are increasingly employed for optical frequency comb stabilization; however, their performance is constrained by the limited control bandwidth. In this work, we present a low-latency digital servo system designed for optical frequency comb stabilization. The system is implemented on a custom-designed FPGA-based digital board. Through the optimization of filtering and phase detection algorithms, computational latency is reduced while maintaining precision, enabling a control bandwidth of 1.1 MHz. The repetition rate of the optical frequency comb is locked to a 1560 nm ultrastable laser. The modified Allan deviation of the measured beat signal, normalized to the optical frequency, is 7.27×10^{-18} at 1 second, thereby validating the performance of the proposed digital servo system.

I. INTRODUCTION

Optical frequency combs are powerful instruments comprising millions of mutually coherent, evenly spaced modes distributed over a broad frequency range^{1,2}. Serving as a bridge between optical and microwave domains^{3,4}, they have demonstrated remarkable application potential in areas such as precision spectroscopy⁵⁻⁷, low-noise microwave generation⁸⁻¹⁰, two-way time-frequency transfer¹¹⁻¹³, atomic clock network¹⁴⁻¹⁶ and laser ranging¹⁷⁻²⁰. In these applications, frequency stability and accuracy are critical parameters²¹.

The frequency modes of an optical comb can be expressed as $f_n = n f_{rep} + f_{ceo}$, where n is an integer typically on the order of $10^5 - 10^6$, f_{rep} is the pulse repetition frequency corresponding to the intermode spacing, and f_{ceo} is the carrier-envelope offset frequency^{22,23}. According to the operating principles of optical combs, two degrees of freedom must be controlled. Frequency stabilization is typically achieved by simultaneously locking the f_{ceo} and the frequency of a selected comb mode (f_n). The f_{ceo} can be detected using an $f - 2f$ interferometer²⁴ and controlled via adjustments to the pump current, whereas f_n is stabilized by locking the heterodyne beat frequency (f_{beat}) between the comb mode and an ultrastable laser. Control of f_{beat} is accomplished by tuning the refractive index and cavity length of the optical resonator. Instability in f_{ceo} leads to a global frequency shift across the entire comb spectrum, while f_{beat} determines whether the comb inherits the frequency stability of the ultra-stable laser. Therefore, high-bandwidth, low-noise locking systems are essential

for stable comb operation.

The analog locking system based on the phase-locked loop (PLL) architecture is well established²¹. In this scheme, a detector converts the optical signal into an electrical signal, which is then combined with a reference signal and fed into a phase detector for phase comparison. The output of the phase detector serves as the error signal, which is processed by a proportional-integral-derivative (PID) controller to generate a feedback signal that drives the actuators of the optical comb. Phase locking of the optical comb is achieved by tuning the system parameters to minimize the error signal.

In recent years, FPGA-based digital locking schemes for optical frequency combs have been proposed. The principle of the scheme is illustrated in Fig. 1. A photodiode converts the optical signal into an electrical signal, which is preprocessed through bandpass amplification, frequency division, and filtering before being sent to the digital locking board. The digital board comprises three main components: ADC, FPGA, and DAC. The reference signal is buffered and employed as the FPGA clock. A DDS IP core generates a signal at a specific frequency, which is mixed with the digitized signal from the ADC to extract the differential phase. A PID controller processes the phase information to generate a feedback signal, which is subsequently converted into an analog signal by the DAC. This analog signal is amplified and applied to drive actuators, such as the pump, electro-optic modulator (EOM), and piezoelectric transducer (PZT), to stabilize the optical frequency comb.

Digital locking addresses the inherent limitations of conventional analog systems²⁵. In particular, the phase range of an analog phase detector is limited to 2π rad, and the sys-

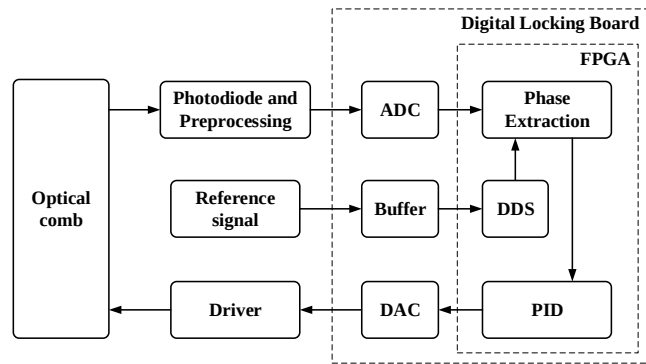


FIG. 1. Block diagram of the digital locking scheme for an optical frequency comb. ADC: analog-to-digital converter; DAC: digital-to-analog converter; DDS: direct digital synthesizer; FPGA: field-programmable gate array.

tem is prone to cycle slipping, which further constrains the capture range of the analog PLL²⁶. By contrast, the digital phase detection module extends the phase range via phase unwrapping and enables slip-free phase locking²⁷. The digital servo system offers high flexibility, as filtering parameters and PID gains can be tuned through software to accommodate various optical frequency combs. It is straightforward to replicate and exhibits enhanced resistance to temperature drift and electromagnetic interference (EMI), thereby ensuring improved stability. Moreover, it integrates advanced diagnostic tools, including built-in frequency counters, spectrum analyzers, and vector network analyzers, which facilitate real-time assessment of the systems locking status.

Nevertheless, the control bandwidth of the digital servo system is constrained by substantial computational latency. Phase detection in digital systems is typically implemented using the IQ quadrature demodulation algorithm²⁸. This demodulation step accounts for the majority of the latency in the digital processing board. For instance, in the widely used Red Pitaya platform, the overall system latency is 407 ns, with 207 ns originating from the demodulation process²⁹. Therefore, algorithmic improvements must strike a balance between computational precision and latency, aiming to reduce latency while preserving sufficient precision.

In this work, we propose a low-latency digital servo and implement an FPGA-based digital locking board. Through optimization of the digital processing algorithm, the demodulation latency is reduced to 50 ns, resulting in a total board latency of less than 250 ns and enabling a control bandwidth of 1.1 MHz. The system was experimentally validated on an erbium-doped fiber optical comb, achieving the expected frequency stability and thereby confirming the practicality of the proposed digital servo.

II. LATENCY OPTIMIZATION

The measurement signal can be expressed as:

$$v_s = A \sin(\omega_s t + \phi_1) \quad (1)$$

By multiplying it with the reference signal and applying filtering, the following expressions are obtained:

$$I = \frac{1}{2} A \sin((\omega_s - \omega_r)t + \phi_1 - \phi_2) \quad (2a)$$

$$Q = \frac{1}{2} A \cos((\omega_s - \omega_r)t + \phi_1 - \phi_2) \quad (2b)$$

The phase difference is then given by:

$$\Delta\phi = \tan^{-1}\left(\frac{I}{Q}\right) = (\omega_s - \omega_r)t + \phi_1 - \phi_2 \quad (3)$$

IQ demodulation involves filtering and arctangent computation, with latency optimization considered for both steps. Finite impulse response (FIR) filters are widely employed in digital filtering owing to their exact linear-phase response and inherent stability; however, they typically require high filter orders, which substantially increase computational complexity. For arctangent computation, the Coordinate Rotation Digital Computer (CORDIC) algorithm is commonly adopted. This iterative algorithm progressively approximates the target value through coordinate rotations, relying solely on simple shift-add operations, which makes it highly suitable for FPGA implementation³⁰. Although it avoids multiplication, the iterative process can still introduce considerable computational latency.

In the locking system of an optical frequency comb, two primary performance metrics are of particular importance: phase detection precision and latency. Both factors critically influence the stability and overall performance of the optical comb, and their impacts will be examined separately in the following sections.

A. Phase Detection Precision

The phase detection precision determines the systems capability to suppress phase noise. Higher phase measurement accuracy enables the system to more effectively detect and correct phase deviations, thereby reducing the linewidth and improving both frequency accuracy and stability. Digital phase detection generally consists of two steps: ADC signal acquisition and subsequent digital signal processing.

In most digital implementations, the ADC sets the ultimate performance of the system³¹. ADC noise arises from multiple sources, including quantization noise, aperture jitter, and reference power supply noise. These noise contributions can be classified into amplitude noise and phase noise³¹. While amplitude noise is largely suppressed during IQ demodulation, phase noise constitutes the dominant source of phase detection error. However, phase noise is rarely specified in ADC datasheets; instead, the signal-to-noise ratio (SNR) is commonly employed as a reference metric. Harmonics and spurious tones introduced by ADC sampling can be suppressed

through digital signal processing, but the residual noise floor still directly affects phase detection precision. The use of low-noise components, such as operational amplifiers, power supplies, decoupling capacitors and appropriate shielding can mitigate electronic noise. Furthermore, an anti-aliasing filter prior to the ADC improves the signal SNR and enhances phase noise performance, while a high-pass filter implemented in the FPGA can suppress low-frequency electromagnetic interference from the environment.

In the digital signal processing stage, additional errors may be introduced by phase truncation of the digital reference signal, insufficient stopband attenuation of the low-pass filter, and numerical truncation in arithmetic operations. A careful trade-off must therefore be made among precision, latency, and hardware resource consumption, ensuring that computational errors remain below the intrinsic phase noise level of the signal and thus do not substantially degrade the overall digital phase detection precision.

B. Latency

Latency refers to the time required for a signal to propagate through the system, and it directly determines the achievable control bandwidth. In the context of optical frequency comb stabilization, the control bandwidth represents the highest frequency at which the system can effectively suppress disturbances or respond to input fluctuations. A lower latency enables the locking system to react to phase deviations more rapidly, thereby increasing the control bandwidth and improving both noise suppression and overall stability.

For the digital locking board, latency can be divided into device latency and processing latency. Device latency arises from the intrinsic delays of hardware components such as the ADC, FPGA, and DAC, and can be minimized by selecting low-latency devices. Processing latency originates from digital signal processing within the FPGA and can be reduced by increasing the FPGA clock frequency or adopting low-latency algorithms. The main digital processing stages include IQ mixing, filtering, arctangent computation, and PID control. IQ mixing and PID primarily involve multiplications and additions, which are implemented with DSP blocks in the FPGA. At high clock frequencies (e.g., 400 MHz), DSP operations typically require three clock cycles to maintain timing stability, whereas at lower frequencies (e.g., 200 MHz), the latency can be reduced to a single cycle. Thus, the FPGA clock frequency must be carefully chosen to balance latency and timing closure. For filtering and arctangent computation, further algorithmic optimizations were implemented to minimize latency, as detailed in the following sections.

1. Filter

As shown in Eqs. (2), after IQ mixing, a low-pass filter is required to suppress the doubled-frequency component generated during mixing as well as potential local oscillator leakage. Simulation results demonstrate that the filtering perfor-

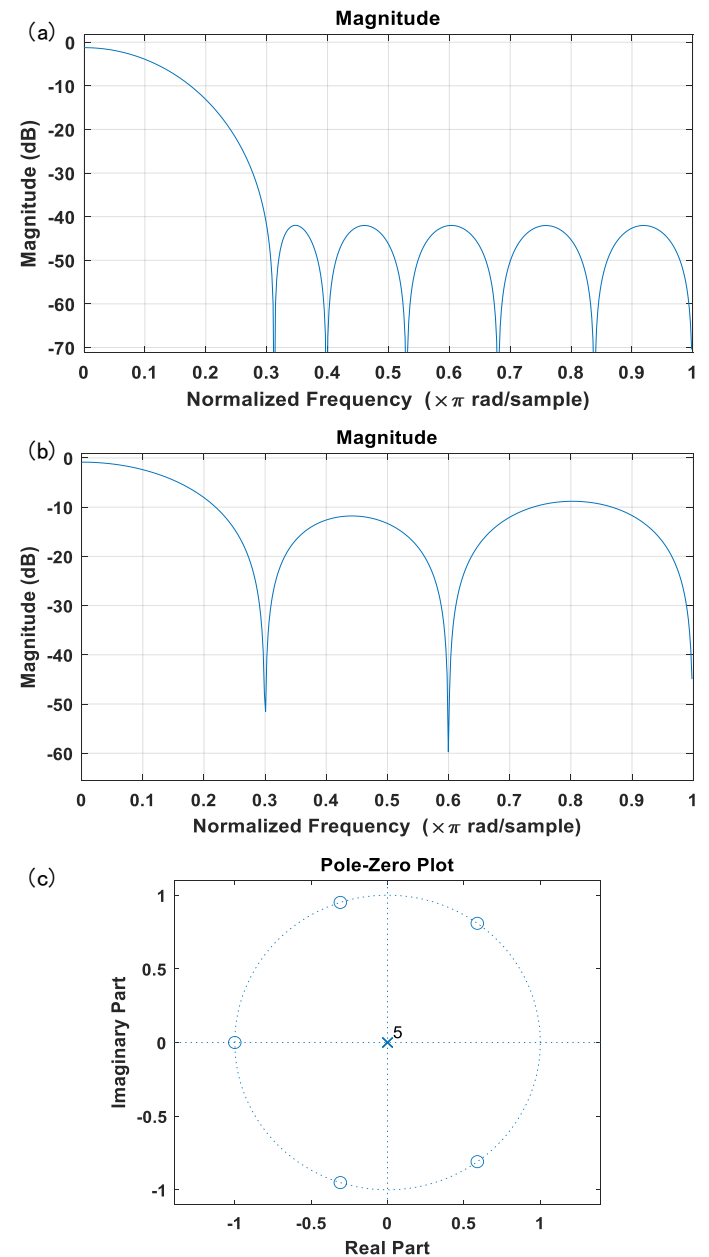


FIG. 2. Magnitude responses of the (a) low-pass filter and (b) notch filter, and (c) the polezero plot of the notch filter.

mance has a significant impact on phase detection: for every 20 dB increase in stopband attenuation, the phase error decreases by approximately one order of magnitude. At a sampling rate of 100 MHz, with a stopband frequency of 15 MHz and a stopband attenuation of 40 dB, a filter designed using the equiripple method requires an 11th-order implementation, as shown in Fig. 2(a), with a group delay of 5.5 sampling periods. A lower cutoff frequency or higher stopband attenuation would necessitate an even higher-order filter. It is worth noting that our objective is to suppress the signal at two specific frequency points, rather than across the entire stopband. The linewidth of the frequency comb is approximately 200 kHz

prior to locking^{21,32}, and can be reduced to the mHz level after locking³³. Owing to this relatively narrow frequency range, employing a notch filter offers a feasible alternative to a conventional low-pass filter. By adding zeros corresponding to the targeted frequencies in the pole-zero design, attenuation within a narrow range around these frequencies can be ensured, with attenuation at the exact frequency points even greater than that of the low-pass filter. The final FIR filter design is of 5th order, as shown in Fig. 2(b) and Fig. 2(c), and contains four conjugate zeros on the unit circle together with one zero at an angle of π , which prevents frequency aliasing. Owing to its lower order, the group delay is reduced to only 2.5 sampling periods. The notch filter is effective only at specific frequency points, and thus requires adjustment if the locking frequency changes. However, in an optical comb locking system, the locking frequency does not need to change frequently. Importantly, the filter order remains constant across different frequencies, and only the positions of the zeros must be modified to update the filter coefficients, enabling rapid adjustment of the filters frequency response. When implemented in an FPGA, the inherent parallelism of the hardware can be exploited to pre-compute partial results of previous samples before the arrival of the next one. Consequently, each filtering step requires only a single multiply-accumulate operation, and the overall latency can be reduced to just four clock cycles.

2. Arctangent Computation

An alternative method using lookup tables (LUTs) for arctangent computation is developed in this paper to reduce latency. The algorithm first classifies the signal according to its sign and magnitude, then converts the IQ ratio into a multiplication with a pre-stored reciprocal, followed by mapping the result to an angle value through another lookup table. The computed value is finally extended to the full range $(-\pi, \pi)$.

Implementing division directly in an FPGA is challenging due to its high latency, but by replacing division with reciprocal multiplication, the processing speed is significantly improved. To further minimize resource usage, only angle values corresponding to positive IQ ratios in the first quadrant are stored. Values for the other quadrants can be obtained through symmetry using addition or subtraction, based on the properties of the arctangent function. Directly storing reciprocals near zero is impractical because of their large dynamic range and poor precision. To overcome this, we use the identity $\tan^{-1}\left|\frac{I}{Q}\right| + \tan^{-1}\left|\frac{Q}{I}\right| = \frac{\pi}{2}$, so only the reciprocal of the larger signal is required. This effectively halves the first quadrant. Assuming the signal amplitude is normalized to 1, this condition implies $\max(I, Q) > \frac{\sqrt{2}}{2}$. For margin, the lookup table stores reciprocals in the range $(\frac{1}{2}, 1)$, and the corresponding arctangent table covers input values in $(0, 1)$, mapping to angles between 0 and $\frac{\pi}{4}$. The full $(-\pi, \pi)$ range is then recovered based on the signs and magnitudes of I and Q .

In practice, this design requires that the input signals satisfy $\max(I, Q) > \frac{1}{2}$. In the optical comb locking system, the

amplitude is relatively stable and can be adjusted by tuning the gain of the preceding FIR filter, which changes its parameters without adding latency. For larger amplitude variations, synchronized scaling of the IQ signals ensures that this condition is met. By constraining the processing to one region and eliminating the sign bit, the reciprocal and arctangent lookup tables are implemented with an address width of 14 bits and a data width of 15 bits. The resulting phase is represented in scaled radians with a resolution of 0.024 mrad and a computation latency of only 11 clock cycles. In comparison, the Xilinx CORDIC IP core requires 24 clock cycles to achieve 18-bit precision.

III. IMPLEMENTATION AND TEST

A digital locking board was designed to implement a dual-loop locking system for the optical frequency comb, namely the f_{ceo} and f_{beat} loops. The f_{ceo} loop provides fast feedback control for the pump, whereas the f_{beat} loop implements fast feedback for the EOM and slow feedback for the PZT. The board is based on a Xilinx 7k325t FPGA and integrates two 16-bit ADCs with a sampling rate of 100 MSPS, two 16-bit DACs with a refresh rate of 50 MSPS, and a high-precision 20-bit DAC with a refresh rate of 1 MSPS. The input stage features modular frequency dividers and low-pass filters to accommodate different locking frequencies, thereby enhancing system flexibility. These modules can reduce the required ADC sampling rate and system bandwidth, increase the phase-locking range, and improve stability, although they also introduce some latency. The board outputs amplified signals through operational amplifiers supporting ± 10 V, which are sufficient to drive the pump and EOM. For scenarios requiring higher PZT voltage, a separate amplification module can be inserted at a subsequent stage to prevent high-voltage noise from affecting the system. A clock input is distributed to all devices via a clock divider and distributor to achieve full-board synchronization. This clock, serving as the reference for the locking system, can originate from a high-stability frequency standard, such as a rubidium clock, or from the repetition frequency signal of the optical comb, enabling self-locking operation. The board interfaces with a host computer via USB, and dedicated software developed in C# enables parameter configuration and real-time monitoring of the system status.

Fig. 3 shows the logic block diagram of the FPGA, operating at a clock frequency of 400 MHz. The digital signal first passes through a first-order IIR high-pass filter with a cutoff frequency of several kHz, designed to require only addition, subtraction, and shift operations to minimize computational latency. The digital reference signal is generated by a direct digital synthesizer (DDS) operating at 100 MHz with a 48-bit frequency control word, yielding a frequency resolution of 0.355 μ Hz. After mixing, the signal is filtered by a 5th-order FIR filter to remove specific frequency components and subsequently processed using division and arctangent operations via lookup tables to compute the phase. The resulting output is an 18-bit number spanning a range of 2π radians. A reg-

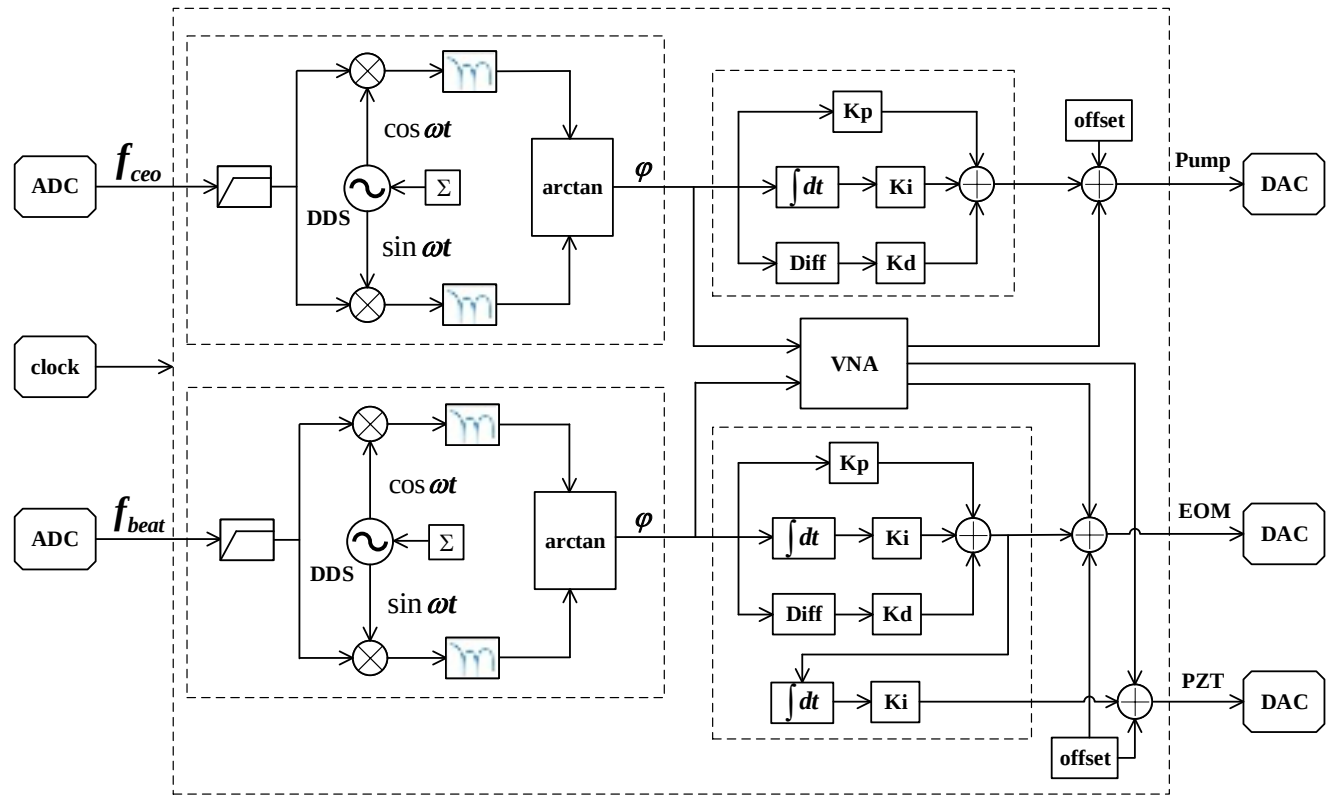


FIG. 3. FPGA logic block diagram. Channel 1 is dedicated to the f_{ceo} loop, and Channel 2 is dedicated to the f_{beat} loop. The Σ symbol denotes the phase accumulator for recording the DDS initial phase. Kp, Ki, and Kd represent the proportional, integral, and derivative gains of the PID controller, respectively.

ister records the cycle, and the phase is unwrapped to extend the range over multiple cycles. Both the cycle and phase serve as error signals for the PID controller, ensuring slip-free locking. While the phase itself is unbounded and may cause output saturation in the P channel, the phase difference remains bounded. The D and P channels are used to lock the phase difference (equivalent to frequency locking), after which the I channel can be enabled to achieve full phase locking. Thus, the system generally operates by first locking the frequency, followed by phase locking. The output signal allows offset adjustment. The slow feedback PID exclusively utilizes the I channel, receiving the output from the fast feedback PID as its input. Cascading the two PIDs expands the frequency adjustment range and enables higher gain at low frequencies. The phase detection result is added to the DDS phase to represent the measured signal phase. By computing differences at fixed time intervals and averaging the resulting frequencies, a frequency counter functionality is realized, which is used to assess the long-term frequency stability of the phase-locked loop. Additionally, the FPGA implements a digital vector network analyzer (VNA) to measure the amplitude and phase frequency response of the loop transfer function, facilitating PID parameter tuning. Under lock, the optical comb phase can be controlled by adjusting the initial DDS phase. A phase accumulator records multi-cycle phase changes, representing a key

advantage of digital over analog servos. Table I summarizes the FPGA resource utilization, indicating sufficient capacity for further optimization of the control algorithms.

TABLE I. FPGA resources utilization.

Resource	Used	Available
Look-up table (LUT)	5494	203800
LUT RAM	400	64000
Flip-flops (FF)	10796	407600
Block RAM (BRAM)	71	445
DSP slices	62	840
Global clock buffer (BUFG)	6	32
Mixed-mode clock manager (MMCM)	2	10

An anti-aliasing filter is placed before the ADC to suppress high-frequency components, with a narrower filter bandwidth reducing the input noise for the ADC. In tests conducted with a 50 MHz bandwidth, the ADC exhibited a signal-to-noise ratio (SNR) of 77 dB, corresponding to an effective resolution of 12.2 bits. To evaluate phase detection accuracy, a test signal derived from the same source as the clock was applied, with a frequency of 15 MHz and an amplitude of 8 dBm, close to the full-scale ADC range. Figure 4 shows the phase calculated by the FPGA, with a standard deviation of 0.093 mrad. In addition, the raw ADC codes were recorded and processed

using the same algorithm on a computer, yielding a standard deviation of 0.090 mrad. The FPGA-calculated phase exhibits a sawtooth pattern due to truncation, whereas the computer-calculated phase introduces negligible error. Comparison between the FPGA and computer results demonstrates that the phase waveforms are highly consistent, with only minor differences in standard deviation. This verifies the feasibility of low-latency phase calculation on the FPGA and indicates that ADC performance is the primary limiting factor in phase detection accuracy. The signal amplitude also significantly affects phase accuracy: at lower amplitudes, fewer ADC codes are used, increasing phase error. Under identical test conditions, when the signal amplitude was reduced to 6 dBm, the FPGA-calculated phase standard deviation increased to 0.111 mrad, while the computer-calculated phase standard deviation increased to 0.106 mrad.

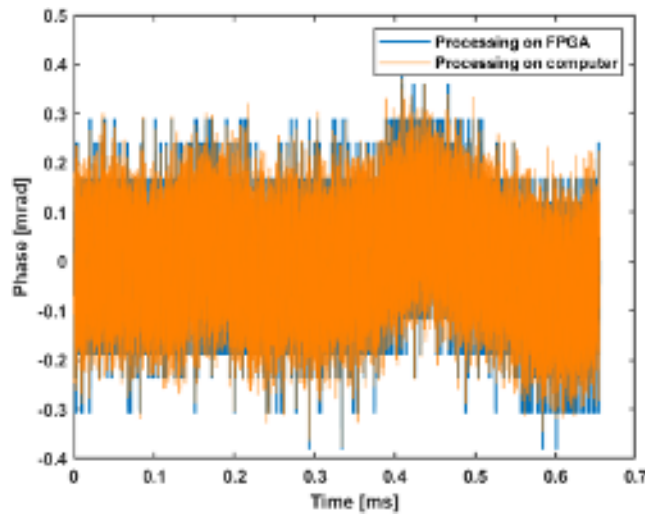


FIG. 4. Phase deviation measured with a 15 MHz, 8 dBm input signal.

The latency of the board can be estimated from the system bandwidth, which is derived from the loops phase-frequency response. To characterize this, the board's output was connected back to its input, forming a closed test loop that was analyzed using the FPGA-implemented digital VNA. In the first configuration, the ADC codes were used as the input and the DAC codes as the output, excluding digital signal processing within the FPGA (i.e., without phase calculation). The resulting phase-frequency response, shown as the first curve in Fig. 5, exhibited a bandwidth of approximately 1.44 MHz, corresponding primarily to the intrinsic device latency. The low-frequency rise of the curve is attributed to the AC-coupled input, which introduces a high-pass effect. Since the board operates as a phase system while the VNA is designed for amplitude systems, a DDS was employed to generate a swept signal that phase-modulates a fixed-frequency carrier. The modulated signal was output and compared with the input signal, enabling the VNA to measure the phase response of the system. In the second configuration, the PID output was taken as the VNA input, thereby incorporating the full digital signal processing chain in the FPGA (i.e., including phase calculation).

The corresponding phase-frequency response, shown as the second curve in Fig. 5, yielded a bandwidth of about 1.1 MHz, from which the overall board latency was estimated at approximately 227 ns.

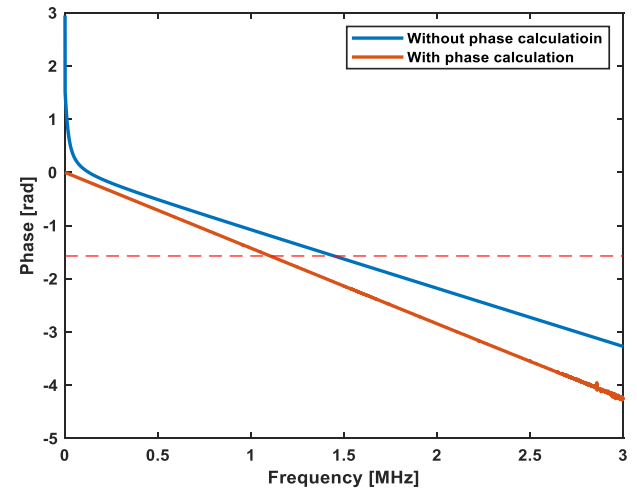


FIG. 5. Phase-frequency response of the digital locking board. The control bandwidth, defined as the frequency at which the phase shift reaches $-\pi/2$ (indicated by the dashed line), is 1.44 MHz without phase calculation and 1.1 MHz with phase calculation.

For comparison, latency parameters from common solutions, such as commercial boards (e.g., Red Pitaya) and customized System on Chip (SoCs)³⁴, are summarized in Table II. The results indicate that our digital servo achieves significantly lower latency, a performance gain attributed to the optimization of the digital processing algorithms.

TABLE II. Latency comparison of digital servo systems.

Work-Year	latency
Red Pitaya-2015 ²⁵	320 ns excluding phase detection (ADC: 70ns, DAC: 70ns)
Red Pitaya-2018 ²⁹	407 ns (phase detection: 207ns)
RFSoc-2025 ³⁵	1.35 μ s
This work-2025	227 ns (ADC: 70ns, DAC: 28ns, phase detection: 50ns)

A locking test was performed on a 100 MHz repetition frequency erbium-doped fiber optical comb, using a 1560 nm ultrastable laser as the reference. The locking frequency was set to 60 MHz with a division factor of four. The spectrum of the locked signal, shown in Fig. 6(a), exhibits narrow-linewidth features, confirming effective stabilization. The actual feedback bandwidth is approximately 200 kHz, which is smaller than the 1.1 MHz servo bandwidth and is limited by the response speed of the optical frequency comb controller. Phase noise measurements of f_{ceo} and f_{beat} , obtained with an AnaPico APPH20G phase noise analyzer, are shown in Fig. 6(b) and Fig. 6(c), covering the 1 Hz - 1 MHz range. The integrated phase noise within this range is 469 mrad for f_{ceo} .

and 168 mrad for f_{beat} , corresponding to timing jitter of 139 attoseconds. These results demonstrate that the digital locking system provides strong suppression of phase noise in the optical comb.

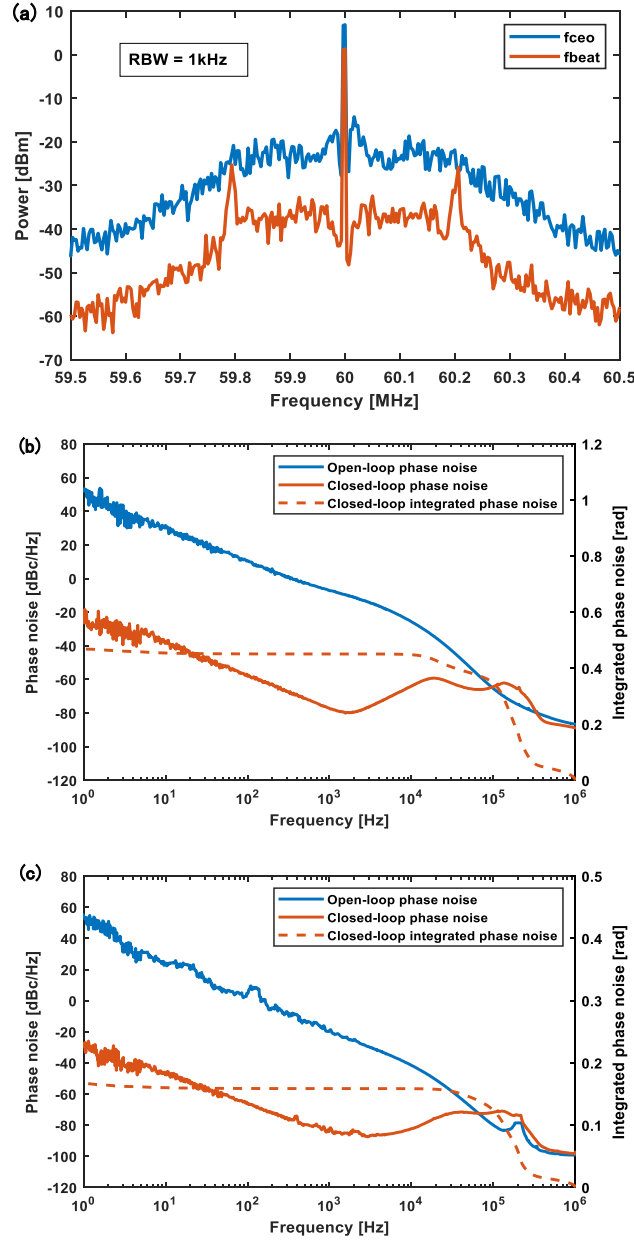


FIG. 6. (a) In-loop RF spectrum of f_{ceo} and f_{beat} . (b) Phase noise and integrated phase noise of f_{ceo} (469 mrad from 1 Hz to 1 MHz). (c) Phase noise and integrated phase noise of f_{beat} (168 mrad from 1 Hz to 1 MHz).

Additionally, the long-term stability of the locked optical comb was evaluated. A K+K Messtechnik FXQE80 frequency counter was used with a 1-s gate time and 1000-point averaging to record the signal frequency continuously for over 10 hours. The modified Allan deviation (MDEV), normalized to the optical reference frequency, is shown in Fig. 7. For f_{ceo} , it reaches 2.16×10^{-17} at 1 s and improves to 1.04×10^{-22} at

10,000 s. In comparison, f_{beat} exhibits 7.27×10^{-18} at 1 s and 1.29×10^{-22} at 10,000 s. These results confirm that the optical comb achieves the expected level of frequency stability.

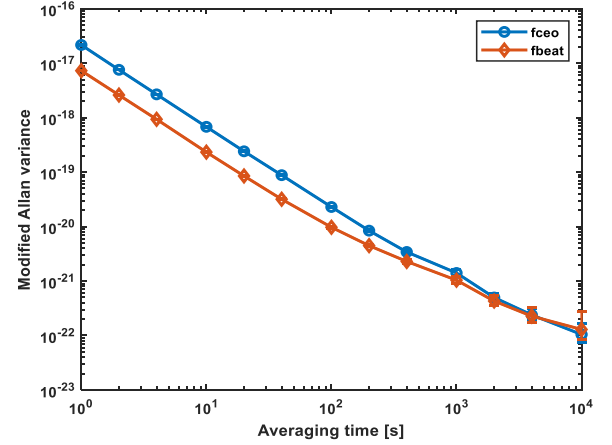


FIG. 7. Modified Allan deviations of the locked optical comb, normalized to the optical reference laser at 1560 nm.

We compared the locking performance of the proposed algorithm with that of a conventional high-order filter and an arctangent computation. The high-order filter introduces a latency of 7 clock cycles, while the arctangent computation requires 24 cycles. This constitutes an additional 16 cycles of delay compared to our proposed algorithm, corresponding to 40 ns at a 400 MHz clock rate. Owing to the bandwidth limitation of the comb drivers, only minor differences in locking performance are observed between the proposed and conventional algorithms. To further evaluate the impact of latency, additional delays were intentionally added to the proposed algorithm. The results show that increasing delay degrades both the locking bandwidth and frequency stability of the comb. The detailed measurements are summarized in Table III.

TABLE III. Comparison of optical comb locking performance.

Design method	Bandwidth	MDEV at 1 s
This work	~100 kHz	f_{ceo} : 2.40×10^{-17} f_{beat} : 1.95×10^{-17}
Normal algorithm	~100 kHz	f_{ceo} : 3.02×10^{-17} f_{beat} : 2.02×10^{-17}
Delay 300 ns	~80 kHz	f_{ceo} : 3.58×10^{-17} f_{beat} : 2.72×10^{-17}
Delay 500 ns	~70 kHz	f_{ceo} : 4.40×10^{-17} f_{beat} : 2.88×10^{-17}

With the increasing deployment of optical frequency combs outside laboratory environments^{36–38}, the stability of the electronics system becomes more critical under complex and variable conditions. Compared with analog locking, digital locking provides superior immunity to interference; however, the ADC remains sensitive to temperature variations. In thermal chamber tests, the phase of a clock-synchronous signal

was observed to drift by -0.225 mrad per degree Celsius, as shown in Fig. 8. Phase-lock tests under temperature variations exceeding 10°C revealed phase fluctuations on the order of milliradians-comparable to the noise floor-with no observable degradation in locking bandwidth or loop gain. This indicates that temperature-induced effects on latency and filter parameters are negligible. Vibrations, whose frequencies typically fall within the locking bandwidth, can be suppressed by the feedback loop unless their amplitude is large enough to cause a loss of lock; additional vibration isolation can also be applied if necessary. Electromagnetic interference can be effectively mitigated through shielding. These results collectively demonstrate that the proposed electronics system enables stable frequency comb operation under complex environmental conditions.

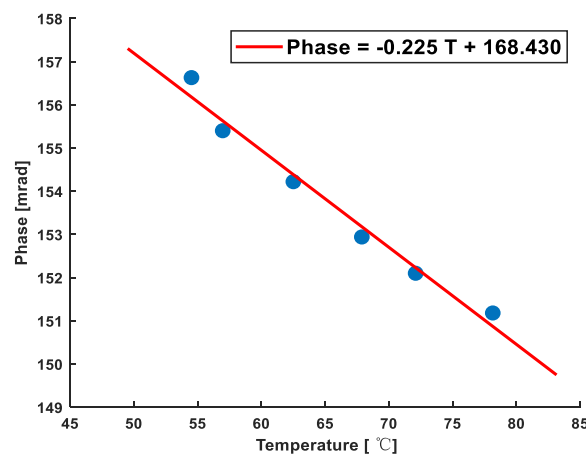


FIG. 8. Measured and fitted phase variation as a function of temperature.

IV. CONCLUSION

We designed an FPGA-based digital servo system with two locking loops, which can be flexibly configured to lock optical frequency comb systems controlled by EOM, PZT, and pump current sources. By improving the filter and phase detection algorithms to minimize computational latency, the system achieves a phase detection precision of 0.11 mrad and a servo bandwidth of 1.1 MHz. In locking tests with an erbium-doped fiber comb, the integrated phase noise of f_{ceo} and f_{beat} was measured as 469 mrad and 168 mrad, corresponding to timing jitter of 139 attoseconds. Cycle-slip free locking lasted for over 10 hours, achieving modified Allan deviations of 2.16×10^{-17} and 7.27×10^{-18} at 1 second for f_{ceo} and f_{beat} . This low-latency digital servo system enables higher locking bandwidth and, due to its low cost, flexible configuration, and scalability, is expected to find broader applications in optical locking.

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DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding authors upon reasonable request.

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